

Research Interests

- Computational Linguistics and Psycholinguistics
- Computational and information-theoretic models of sentence processing
- Syntactic theories and parsing

Awards and Honors

Director's Award National Institute for Japanese Language and Linguistics (NINJAL) Institute-wide award recognizing outstanding research achievements	2025
Young Researcher Encouragement Award National Institute for Japanese Language and Linguistics (NINJAL) Awarded to early-career researchers for outstanding research achievements	2025
Best Paper Award Conference on Computational Natural Language Learning (CoNLL) <i>Is Structure Dependence Shaped for Efficient Communication?: A Case Study on Coordination</i>	2024
Young Researcher Encouragement Award The Association for Natural Language Processing Annual Conference in Japan Awarded to early-career researchers for outstanding research achievements	2025
Committee Special Award The Association for Natural Language Processing Annual Conference in Japan Awarded for exceptional research contributions in Japanese NLP (received in consecutive years)	2024, 2025

Education

Georgetown University Ph.D. Student Department of Linguistics Advisor: Prof. Ethan Gotlieb Wilcox	Washington, D.C., USA August 2025 – May 2030 (expected)
The Graduate University for Advanced Studies, SOKENDAI Ph.D. Student National Institute for Japanese Language and Linguistics (NINJAL) Advisor: Prof. Yusuke Kubota	Tokyo, Japan April 2025 – August 2025
The University of Tokyo M.A. in Arts and Sciences (Linguistics) Department of Language and Information Sciences Thesis: <i>Structure Dependence is Shaped for Efficient Communication</i>	Tokyo, Japan April 2023 – March 2025
The University of Tokyo B.A. in Arts and Sciences (Major: Linguistics, Minor: Computer Science) College of Arts and Sciences Thesis: <i>Incremental Semantic Composition as Psycholinguistically Plausible Strategy</i>	Tokyo, Japan April 2019 – March 2023
Additional Training Summer School on Statistical Methods for Linguistics and Psychology University of Potsdam Advanced Bayesian data analysis	Potsdam, Germany August 2026

Publications

Preprints

1. **Kohei Kajikawa** and Shinnosuke Isono. [The Dual Nature of Syntactic Node Count: Facilitating and Inhibiting Sentence Comprehension](#). *PsyArXiv: preprint*.

Journal Articles

1. Shinnosuke Isono, **Kohei Kajikawa**, Yohei Oseki, and Masayuki Asahara. [Modeling Memory Effects in a Head-final Language with Category Locality](#). *Open Mind*, 2026. 10: 1359–1393.
2. Kai Nakaishi, Ryo Yoshida, **Kohei Kajikawa**, Koji Hukushima, and Yohei Oseki. [Rethinking the Relationship between the Power Law and Hierarchical Structures](#). *Transactions of the Association for Computational Linguistics (TACL)*, 2026. 14: 1917–1935.

Refereed Conference Proceedings

1. **Kohei Kajikawa***, Lin Ai*, Tatsuki Kuribayashi, and Ethan Gotlieb Wilcox. [Using LMs to Model the Effects of Context and Coreference during Sentence Comprehension](#). *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.
2. **Kohei Kajikawa**, Shinnosuke Isono, and Ethan Gotlieb Wilcox. [Information-Theoretic Storage Cost in Sentence Comprehension](#). *Proceedings of the 30th Conference on Computational Natural Language Learning (CoNLL)*, 2026. 15–33.
3. Shinnosuke Isono and **Kohei Kajikawa**. [Syntactically-guided Information Maintenance in Sentence Comprehension](#). *Proceedings of the 30th Conference on Computational Natural Language Learning (CoNLL)*, 2026. 57–69.
4. Ryo Yoshida, Shinnosuke Isono, **Kohei Kajikawa**, Taiga Someya, Yushi Sugimoto, and Yohei Oseki. [If Attention Serves as a Cognitive Model of Human Memory Retrieval, What is the Plausible Memory Representation?](#). *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025. 9795–9812.
5. **Kohei Kajikawa**, Yusuke Kubota, and Yohei Oseki. [Is Structure Dependence Shaped for Efficient Communication?: A Case Study on Coordination](#). *Proceedings of the 28th Conference on Computational Natural Language Learning (CoNLL)*, 2024. 291–302. [🏆 Best Paper Award]
6. **Kohei Kajikawa**, Ryo Yoshida, and Yohei Oseki. [Dissociating Syntactic Operations via Composition Count](#). *Proceedings of the 46th Annual Meeting of the Cognitive Science Society (CogSci)*, 2024. 297–305.
7. Shinnosuke Isono, Takuya Hasegawa, **Kohei Kajikawa**, Koichi Kono, Shiho Nakamura, and Yohei Oseki. [Formalizing Argument Structures with Combinatory Categorical Grammar](#). *Lecture Notes in Computer Science*, 2023. 83–99.

Invited Articles

1. **Kohei Kajikawa**. [Exploring New Frontiers in Computational Psycholinguistics: Testing the Efficient Communication Hypothesis](#). *Journal of Natural Language Processing*, 2025. (written in Japanese)

Presentations

Conference Presentations

1. **Kohei Kajikawa***, Lin Ai*, Tatsuki Kuribayashi, and Ethan Gotlieb Wilcox. [Using LMs to Model the Effects of Context and Coreference during Sentence Comprehension](#). Oral presentation at *the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Budapest, Hungary, 2026.
2. Yushi Sugimoto*, **Kohei Kajikawa***, Shinnosuke Isono*. [Information-theoretic storage cost in the brain: fMRI and EEG evidence from naturalistic sentence comprehension](#). Poster presentation at *Society for the Neurobiology of Language (SNL 2026)*, Zurich, Switzerland, 2026.
3. **Kohei Kajikawa**, Shinnosuke Isono, and Ethan Gotlieb Wilcox. [Information-Theoretic Storage Cost in Sentence Comprehension](#). Oral presentation at *the 30th Conference on Computational Natural Language Learning (CoNLL)*, San Diego, CA, USA, 2026.
4. Shinnosuke Isono and **Kohei Kajikawa**. [Syntactically-guided Information Maintenance in Sentence Comprehension](#). Oral presentation at *the 30th Conference on Computational Natural Language Learning (CoNLL)*, San Diego, CA, USA, 2026.
5. **Kohei Kajikawa** and Ethan Gotlieb Wilcox. [Spillover as Rational Processing Delay in Sentence Comprehension](#). Poster presentation at *Annual Meeting of Human Sentence Processing (HSP 2026)*, Cambridge, MA, USA, 2026.
6. **Kohei Kajikawa** and Shinnosuke Isono. [Syntactic Node Count as Index of Predictability](#). Poster presentation at *Computational Psycholinguistics Meeting (CPL 2025)*, Utrecht, Netherlands, 2025.
7. Shinnosuke Isono and **Kohei Kajikawa**. [Grammar as logic, processing as deduction, actions as theorems, states as propositions](#). Poster presentation at *Computational Psycholinguistics Meeting (CPL 2025)*, Utrecht, Netherlands, 2025.
8. Shinnosuke Isono*, **Kohei Kajikawa***, Yushi Sugimoto*, Masayuki Asahara, and Yohei Oseki. [Exploring spatial and temporal dynamics of language comprehension in the brain with CCG](#). Poster presentation at *Annual Meeting of Cognitive Science Society (CogSci 2025)*, San Francisco, CA, USA, 2025.
9. **Kohei Kajikawa**, Yusuke Kubota, and Yohei Oseki. [Is Structure Dependence Shaped for Efficient Communication?: A Case Study on Coordination](#). Oral presentation at *the 28th Conference on Computational Natural Language Learning (CoNLL)*, Miami, FL, USA, 2024.
10. **Kohei Kajikawa**, Ryo Yoshida, and Yohei Oseki. [Dissociating Syntactic Operations via Composition Count](#). Oral presentation at *the 46th Annual Meeting of the Cognitive Science Society (CogSci)*, Rotterdam, Netherlands, 2024.
11. **Kohei Kajikawa**. [Analyzing Japanese Cleft Construction in Combinatory Categorical Grammar](#). Oral presentation at *Logic and Engineering of Natural Language Semantics (LENLS 2023)*, Osaka, Japan, 2023.
12. Shinnosuke Isono, Takuya Hasegawa, **Kohei Kajikawa**, Koichi Kono, Shiho Nakamura, and Yohei Oseki. [Formalizing Argument Structures with Combinatory Categorical Grammar](#). Oral presentation at *Logic and Engineering of Natural Language Semantics (LENLS 2022)*, Tokyo, Japan, 2022.

Invited Presentations

1. An Information-Theoretic Approach to Human Sentence Processing. The 303rd NINJAL Salon. National Institute for Japanese Language and Linguistics, July 14, 2026.
2. Is Structure Dependence Shaped for Efficient Communication?: A Case Study on Coordination. NLP colloquium. Online, Jan 29, 2025.

Grants and Fellowships

HUMAI Program Grant B	2026–2027
JPY 700K (research grant)	
Supported by Matsuo-Iwasawa Lab UTokyo/The Nippon Foundation	
Travel Grant for 39th Annual Conference on Human Sentence Processing	2026
Supported by Human Sentence Processing	
Travel Grant for Computational Psycholinguistics Meeting 2025	2025
Supported by MultipleYEY COST Action	
Travel Grant for Computational Psycholinguistics Meeting 2025	2025
Supported by the Department of Linguistics, Georgetown University	
JSPS Research Fellowships for Young Scientists (DC1)	2025
JPY 200K/month + JPY 3.9M (research grant over 3 years)	
Declined the fellowship upon accepting an offer to pursue doctoral training abroad.	

Research Positions

Research Assistant	Georgetown University
Supervisor: Prof. Ethan Gotlieb Wilcox	August 2026 – December 2026
Research Intern	NTT Communication Science Laboratories
Mentor: Dr. Sanae Fujita	July 2026 – August 2026
Research Assistant	National Institute of Informatics (NII)
Supervisor: Prof. Saku Sugawara	June 2026 – July 2026
Research Assistant	The University of Tokyo
Supervisor: Prof. Yoshifumi Kawasaki	June 2026 – July 2026
Research Assistant	Georgetown University
Supervisor: Prof. Nathan Schneider	January 2026 – April 2026
Research Assistant	Georgetown University
Supervisor: Prof. Ethan Gotlieb Wilcox	August 2025 – December 2025
Research Assistant	National Institute for Japanese Language and Linguistics (NINJAL)
Supervisor: Prof. Yusuke Kubota	May 2025 – August 2025
Research Fellow (DC1)	SOKENDAI
Advisor: Prof. Yusuke Kubota	April 2025 – August 2025
Technical Staff	National Institute for Japanese Language and Linguistics (NINJAL)
Supervisor: Prof. Yusuke Kubota	June 2023 – May 2025

Teaching Experience

Teaching Assistant	The University of Tokyo
<i>Introduction to Natural Language Processing</i> (College of Arts and Sciences)	October 2024 – March 2024
<i>Language of Humans and AI</i> (College of Arts and Sciences)	April 2024 – September 2024
<i>Linguistics as Cognitive Science</i> (College of Arts and Sciences)	April 2023 – September 2023

Academic Service

Reviewer for CoNLL2025 (outstanding reviewer), LREC2026, CMCL2026, CoNLL2026, CPL2026, ARR2026